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Multi-Grade Diabetic Retinopathy Classification using Feature-Fused Hybrid AI Models

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***Abstract*— Diabetic retinopathy (DR) is a leading cause of vision loss, often going unnoticed until irreversible damage is done. Early detection is critical for preventing blindness, but manual grading by ophthalmologists is time-consuming and error-prone. The proposed system proposes a hybrid DL-ML model for multi-grade DR classification using retinal fundus images. Three pre-trained CNNs, DenseNet-121, ResNet-50, and EfficientNet-B3, were developed to extract complimentary visual features. DenseNet-121 prioritizes feature reuse, ResNet-50 improves deep feature learning with residual connections, and EfficientNet-B3 balances accuracy and computational economy. Their feature vectors are combined into a single representation and classified with XGBoost, a highly accurate model of complex data interactions. The proposed model has an overall accuracy of 96.49% and class-specific precision ranging from 94.7% for Severe DR to 97% for No DR, outperforming individual CNN benchmarks. By automating DR detection, the system reduces diagnostic delays and human error, allowing for timely intervention to save vision. The findings show that the hybrid framework has the potential to be a clinically relevant decision-support tool, with future options for lightweight deployment and lesion-level interpretability in healthcare settings.**

Keywords*— *Diabetic Retinopathy, Deep Learning, CNN, XGBoost, Feature Fusion, Medical Imaging

I. INTRODUCTION

Diabetic Retinopathy (DR) is a major complication of diabetes that leads to permanent vision loss if not detected early. DR is the leading cause of blindness worldwide, primarily affecting people with high blood sugar. Mild non-proliferative DR causes visible blood vessel damage. Proliferative DR, where new, abnormal blood vessels form, causes vision impairment or blindness. Preventing blindness requires early DR diagnosis and treatment, which can slow or reverse disease damage [1]. Retinal fundus photography is a key DR diagnostic tool. This image shows microaneurysms, haemorrhages, and exudates, which indicate disease progression. DR begins with microaneurysms and progresses to exudates and haemorrhages. Skilled ophthalmologists must manually identify lesions and grade DR severity, which is time-consuming, subjective, and error-prone. Due to the rising global prevalence of diabetes, an efficient, automated system to detect and classify the grades of DR is needed [2]. Deep learning models allow automation of this process, enabling faster, more accurate diagnoses.

The proposed model's objective is to use deep learning to automatically classify diabetic retinopathy. The model will differentiate five types of DR: none, mild, moderate, severe, and proliferative. Microaneurysms, exudates, haemorrhages, cotton wool spots, and neovascularization all contribute to the severity of DR. Given manual grading's limitations, we need automated detection. Manual DR detection takes longer and is more error-prone due to retinal image quality. In areas with a shortage of or overburdened healthcare professionals, automating lesion identification and classification would speed up and improve diagnosis [3]. Machine learning can improve reliability and scalability. Convolutional Neural Networks (CNNs), widely used in medical imaging for feature extraction [1], can extract complex retinal patterns. DR detection can be robust with these models and advanced classifiers like XGBoost. Reusing pretrained models in transfer learning [2] improves performance by reducing training time and increasing accuracy.

Some recent studies have emphasized the importance of hybrid models in DR classification. CNNs combined with machine learning classifiers such as SVM and random forests improve model interpretability and flexibility, allowing healthcare professionals to trust and understand system predictions [4]. Test-Time Augmentation techniques generate multiple augmented test images and combine the results for more stable predictions, improving model robustness [5]. The proposed system describes a novel automated DR classification system that uses machine learning and deep learning to provide an accurate, understandable, and scalable early detection solution. The proposed system combines complementary CNN features, an advanced ensemble classifier, and robust augmentation strategies to address key issues with existing methods, such as limited sensitivity to early-stage DR, class imbalance, and poor interpretability. This design improves the reliability of automated diagnosis and provides clinicians with a useful and dependable tool for making accurate and timely decisions. The system can improve clinical decision-making and patient outcomes better and lower the workload of staff, and make assessments faster and more consistently.

The following describes the paper's organization: Section 2 reviews the relevant literature on the classification of diabetic retinopathy grades. Section 3 discusses the proposed methodology in detail. Section 4 presents the results, and Section 5 concludes the overall work.

II. LITERATURE SURVEY

Many researchers have proposed deep learning and hybrid methods for automatically classifying diabetic retinopathy. Each method sought to improve diagnostic accuracy, decrease computational burden, and enhance clinical interpretability. However, issues like suboptimal per-class performance, dataset

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imbalance, high computational complexity, and poor generalizability persist.

W. Zhou et al. [6] proposed Superpixel Multi-Feature Classification (SMFC) for the detection of retinal fundus exudates. To reduce false positives, they divided images into superpixels and extracted 20 multi-channel intensity and contextual features for each candidate before performing Fisher Discriminant Analysis and optic disc detection. The method had 87.12% sensitivity and 94.01% specificity for DIARETDB1 and e-Ophtha EX. Its scalability was limited by small exudate regions and handcrafted elements.

X. Guo et al. [7] proposed an Enhanced Multi-Feature Fusion Network (EMFN) built on a YOLOv3 backbone and featuring spatial and channel-specific attention modules. The method improved hard exudate detection across different sizes by combining several input features (green channel, CLAHE, morphological, and curvature-based). On the MESSIDOR, HEI-MED, and e-Ophtha EX datasets, the model achieved 93-94% accuracy, but its architecture was not lightweight, and training was computationally expensive due to feature redundancy.

Yasashvini R. et al. [8] investigated hybrid transfer learning models that used CNN, ResNet, and DenseNet for classifying DR into 5 stages. On the Kaggle APTOS dataset, the hybrid CNN-DenseNet achieved 96.22% accuracy, outperforming CNN-ResNet (93.18%) and plain CNN (75.61%). Regardless, class imbalance caused variations in per-class sensitivity and precision.

M. Youldash et al. [9] proposed a multi-model framework built on the EfficientNet, EfficientNetV2, and RegNet architectures. On the APTOS dataset, EfficientNetB3 achieved 85.1% multi-class accuracy, while RegNetX080 achieved 98.6% binary accuracy. Multiple classes' performance was hampered by overlapping features in adjacent DR grades.

L. Arora et al. [10] developed an EfficientNetB0-based model with 86.53% accuracy across 35,108 images. While generalizable and efficient, its performance lagged that of more complex ensembles. K. Rezaee and F. Farnami [11] proposed a hybrid CNN-Transformer model that used both local and global features, with 94.28% and 95.23% accuracy on the APTOS and IDRiD datasets, respectively. Despite its robustness and generalizability, it encountered novel computational and interpretability challenges.

R. Reguant et al. [12] studied CNN interpretability on EyePACS and DIARETDB1, achieving 89-95% accuracy and AUCs of 0.95-0.98. They noticed an overreliance on non-pathological regions, which sparked trust concerns. Hybrid CNN-SVM, boosting, lightweight networks, attention-driven models, GAN augmentation, and explainable AI have improved robustness and transparency [1], [4], [13]. However, poor modularity, overfitting, and inconsistent per-class performance persisted [4].

Despite a lot of advancements in automated diabetic retinopathy research, current methods still have low sensitivity to early-stage DR, poor performance on imbalanced data, limited explainability, and scalability issues. The proposed system fills these gaps by introducing a novel and interpretable pipeline that combines complementary CNN features, integrates an XGBoost classifier, and includes test-time augmentation and feature importance visualization. This approach not only advances the current state of the art, but it also provides a useful and dependable tool for doctors to make accurate and confident predictions in clinical settings.

III. PROPOSED SYSTEM

The proposed system uses a multi-stage pipeline to classify diabetic retinopathy.

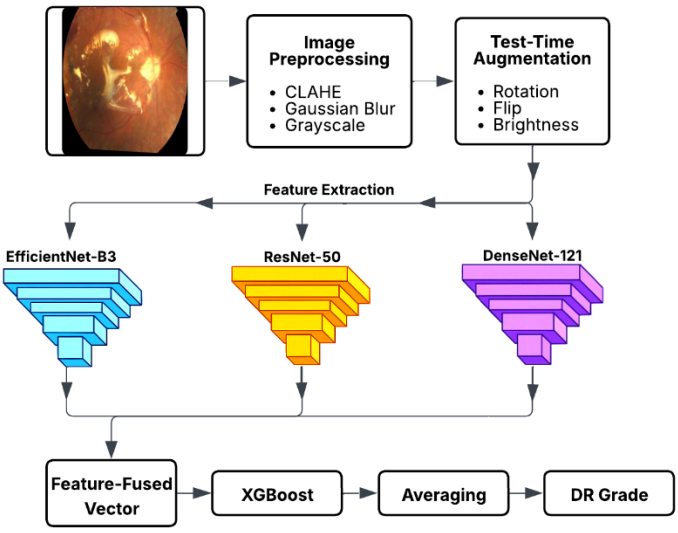

Fig. 1. Execution flow of the Proposed System

Initially, fundus images are preprocessed with CLAHE to improve contrast, Gaussian blur to reduce noise, and grayscale normalization. To improve generalization, Test-Time Augmentation rotates, flips, and changes the brightness. As shown in Figure 1, three deep CNNs are used in parallel to extract features from the retinal fundus images: EfficientNet-B3, ResNet-50, and DenseNet-121, each providing complementary image representations. These feature vectors are concatenated to form a fused vector that captures a variety of distinct patterns. Finally, the fused vector is classified using an XGBoost model, and TTA variant predictions are averaged to determine the final DR grade, ensuring accuracy, efficiency, and clinical interpretability.

A. Image Processing

Image pre-processing improves image quality and makes lesions more visible.
CLAHE (Contrast Limited Adaptive Histogram Equalization) equalizes intensity histograms in small grid regions, improving local contrast and detecting microaneurysms. Next, Gaussian blurring reduces image noise and smoothens out irrelevant high-frequency details, allowing the focus to be on retinal structures. Finally, images are grayscaled to simplify input while preserving spatial patterns. Standardized 384x384 resolution and normalization to [0,1] ensure consistency among CNN models. Figures 2a–2c show images after pre-processing.

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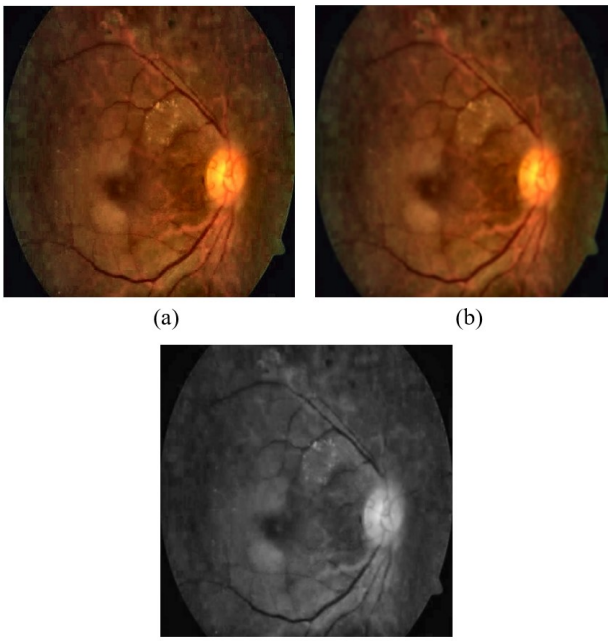

(a) (b)

(c)

Fig. 2. (a) CLAHE, (b) Gaussian Blur, (c) Grayscale

B. Test Time Augmentation

To improve robustness and generalization, the system employs Test-Time Augmentation. During inference, multiple augmented versions of each test image are generated using horizontal flips, slight rotations for $\pm$ 15°, and brightness changes. Each augmented image I_j is passed through the feature extraction pipeline, and their outputs are averaged to obtain a stable representation:

$$F_{\mathrm{TTA}}(I) = \frac{1}{n}\sum_{j=1}^{n} F_j(I) \tag{1}$$

where $F_j(I)$ is the fused feature vector for the jth augmentation. n takes the value 3, which represents the number of augmented versions of each image. This averaged vector is used to make the final prediction, ensuring consistent output across a wide range of input variants. This method is useful for noisy or poorly illuminated test images.

C. Deep Feature Extraction using CNNs

Three pretrained CNN architectures, EfficientNet-B3, ResNet-50, and DenseNet-121, were trained with ImageNet weights and fine-tuned for feature extraction using the pre-processed DR dataset. The final dense classification layers were replaced with global average pooling layers, yielding 1536-, 2048-, and 1024-dimensional feature vectors. The extracted feature vectors for image I are $f_E(I), f_R(I)$, and $f_D(I)$, respectively denoting the 3 models, EfficientNet-B3, ResNet-50, and DenseNet-121.

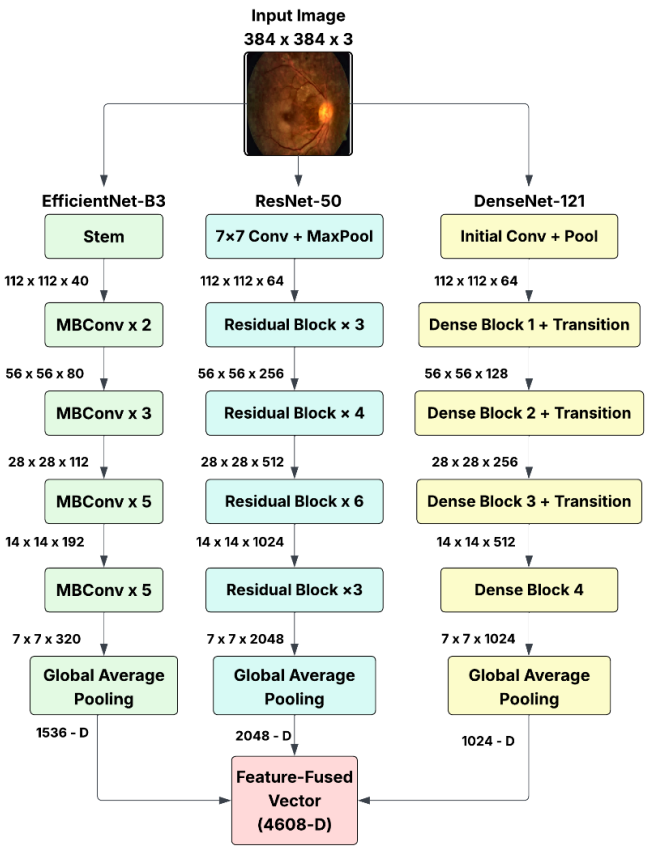

Fig. 3. Architecture of the feature extraction models

Each model had to undergo two training phases: frozen-base and fine-tuning. During the frozen phase, the classification head was trained for 10 epochs using frozen CNN layers and the Adam optimizer with a learning rate of 1×10^{-3}. To fine-tune each CNN, the final 100 layers were unfrozen and trained for 15 epochs at a lower learning rate of 1×10^{-5}. In order to account for class imbalance, a weighted categorical cross-entropy loss function was used, with coefficients of 0.75, 0.81, 0.79, 1.88, and 1.58 for No DR, Mild, Moderate, Severe, and Proliferative. All models had a batch size of 32, and the learning rate was reduced on a plateau and stopped early to prevent overfitting. After training, the model's penultimate layer outputs were concatenated for downstream fusion. This multi-architecture setup combines EfficientNet-B3, ResNet-50's residual learning, and DenseNet-121's dense connections to generate complementary feature representations for DR lesion detection across a range of imaging conditions. Figure 3 shows the architecture of the deep feature extraction models used.

D. Feature Fusion

To combine the strengths of each CNN, the feature vectors are concatenated and fused into one high-dimensional representation.

$$F(I) = [\ f_E(I)\ \|\ f_R(I)\ \|\ f_D(I)\ ] \tag{2}$$

where || denotes vector concatenation.
This fusion strategy produces a consolidated feature vector $F(I) \in \mathbb{R}^d$, which encapsulates a wide array of visual features acquired by the individual CNNs. The fusion enhances representation and mitigates the limitations of individual

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models by integrating their perspectives. For instance, one model may emphasize blood vessel structures, whereas another may more effectively represent texture variations or micro-lesions. Redundancy and diversity improve embedding quality, noise resilience, and XGBoost classifier bias.

E. XGBoost Classifier

The fused feature vector $F(I)$ is then passed into an XGBoost classifier, a gradient boosting framework that constructs additive decision trees in a stage-wise fashion. To avoid overfitting, XGBoost models complex, non-linear visual feature-DR grade relationships with regularization. Its optimization objective is given by $\mathcal{L}(\phi)$.

$$\mathcal{L}(\phi) = \sum_{i=1}^{n} l(y_i, \hat{y}_i) + \sum_{k=1}^{K} \Omega(f_k) \quad (3)$$

where $l(y_i, \hat{y}_i)$ is the classification cross-entropy loss, and $\Omega(f_k)$ penalizes the tree complexity. In this configuration, the maximum tree depth was set to 9, the learning rate was set to 0.03, and there were 500 estimators. The subsampling and feature sampling ratios were both set to 0.8 to improve generalization. The regularization parameters were γ=1 (minimum split loss), λ=2 (L2 regularization on weights), and α=1 (L1 regularization on weights). The classifier returns one of five DR grades.

IV. RESULTS AND DISCUSSIONS

This section evaluates the diabetic retinopathy grade classification model – individual models and the feature-fused model with XGBoost as well. Accuracy, precision, recall, F1 score, and confusion matrix were used to evaluate the models. Jupyter Notebook was used for training, evaluation, and analysis on a Windows 64-bit OS system. The system's NVIDIA RTX 3050 GPU sped up training, making it efficient at handling a large fundus image dataset.

A. Dataset

The system is trained and evaluated on a combination of publicly available datasets that include APTOS, EyePACS, and Messidor, which contain high-resolution retinal fundus images that are labeled into five DR grades. To reduce class imbalance, images from underrepresented classes were augmented and merged across datasets. The final dataset is divided into training, validation, and testing subsets using stratified sampling to keep the class distribution the same across all splits. All images were resized to a uniform size of 384×384 pixels to maintain compatibility with deep CNN architectures.

B. Performance Evaluation of EfficientNet-B3 Model

EfficientNet-B3 showed 64% overall accuracy. Its performance declined for more complex DR grades, including Proliferative DR, even though it performed rather well for No DR with a precision of 0.60, recall of 0.68 and Mild DR.

TABLE I. EfficientNet-B3 Performance Metrics

Class	Accuracy	Precision	F1 Score	Recall	Support
No DR	0.6	0.68	0.64	0.68	2900
Mild DR	0.56	0.63	0.59	0.63	1840
Moderate DR	0.55	0.67	0.6	0.67	2500
Severe DR	0.62	0.65	0.63	0.65	1000
Proliferative DR	0.69	0.57	0.62	0.57	1468

The slightly lower F1-score of the model for Severe DR and Proliferative DR in Table 1 suggests difficulties in accurately identifying these grades.

Fig. 4. Model Accuracy – EfficientNet-B3

The EfficientNet-B3 model shows gradual improvement over epochs, but its training accuracy fluctuates, especially in the earlier stages. The curves in Figure 4 could indicate overfitting or difficulty in handling certain DR grades.

C. Performance Evaluation of ResNet-50 Model

With an overall accuracy of 81%, ResNet-50 did significantly better than EfficientNet-B3. Especially for more difficult grades like Severe DR, this model displayed constant improvement in accuracy, recall, and F1-score as shown in Table II.

TABLE II. ResNet-50 Performance Metrics

Class	Accuracy	Precision	F1 Score	Recall	Support
No DR	0.72	0.83	0.77	0.83	2900
Mild DR	0.71	0.74	0.72	0.74	1840
Moderate DR	0.68	0.81	0.74	0.81	2500
Severe DR	0.78	0.79	0.78	0.79	1000
Proliferative DR	0.84	0.8	0.82	0.8	1468

Given its excellent Severe DR recall, the model's generalizing power exceeded that of EfficientNet-B3.

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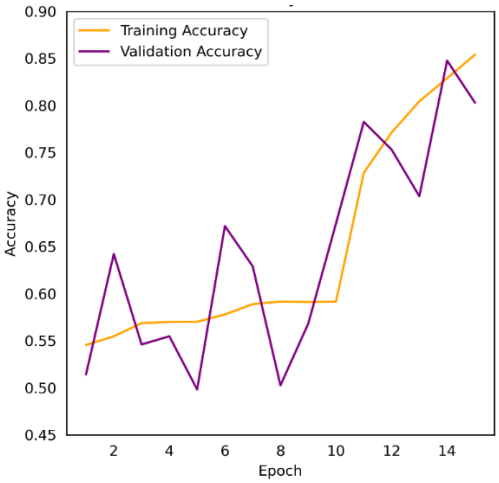

Fig. 5. Model Accuracy – ResNet-50

ResNet-50 performs better in terms of stability during training, with fewer fluctuations compared to EfficientNet-B3. The steady improvement of training accuracy in Figure 5 indicates that the model is converging effectively on the dataset.

D. Performance Evaluation of DenseNet-121 Model

DenseNet-121 outperformed the other two models with an overall accuracy of 94%. As seen in Table III, DenseNet-121 performed consistently well across all grades, particularly in No DR and Severe DR detection.

TABLE III. DenseNet-121 Performance Metrics

Class	Accuracy	Precision	F1 Score	Recall	Support
No DR	0.85	0.88	0.86	0.88	2900
Mild DR	0.77	0.74	0.75	0.74	1840
Moderate DR	0.78	0.82	0.8	0.82	2500
Severe DR	0.8	0.89	0.84	0.89	1000
Proliferative DR	0.85	0.84	0.84	0.84	1468

Its precision, recall, and F1-score were the highest, making it the most accurate model among the 3 stand-alone models.

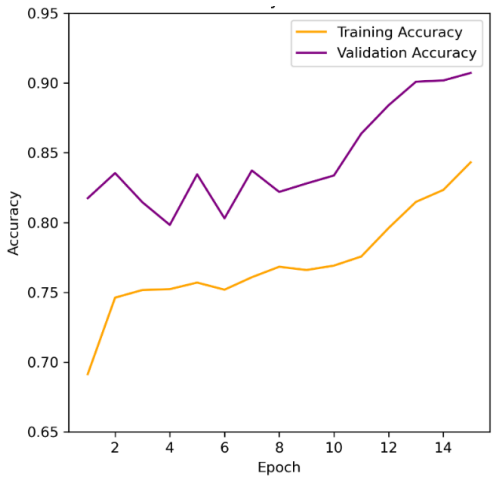

Fig. 6. Model Accuracy – DenseNet-121

Figure 6 is the graph for DenseNet-121, and it displays the most consistent and rapid improvement in training accuracy. The model reaches higher accuracy values compared to the others and converges faster, highlighting its effective feature extraction and generalization capabilities.

E. Proposed feature-Fused Hybrid Model

XGBoost is the final classifier combining the EfficientNet-B3, ResNet-50, and DenseNet-121 models using the feature vector, which was created by fusing the features of the three CNN architectures. This concatenation of features raises general accuracy and helps the model to detect minor trends across all DR classifications. Improving generalization and including features obtained from CNN models depend on the XGBoost algorithm. Table IV shows that the proposed model outperforms the stand-alone models.

TABLE IV. Proposed Hybrid Model's Performance Metrics

Class	Accuracy	Precision	F1 Score	Recall	Support
No DR	0.97	0.97	0.97	0.97	500
Mild DR	0.9722	0.9642	0.9682	0.9642	363
Moderate DR	0.9599	0.9681	0.964	0.9681	470
Severe DR	0.9474	0.9474	0.9474	0.9474	190
Proliferative DR	0.9665	0.963	0.9647	0.963	270
Overall	**96.49%**				

The ROC curve in Figure 7 and corresponding AUC values from Table V indicate that the model rather successfully distinguishes among several DR grades. The model shows good performance in all grades; the Moderate DR grade shows the best AUC, indicating its increased capacity to distinguish between Moderate DR and other categories.

TABLE V. Proposed Hybrid Model – AUC Values

DR Class	AUC
No DR	0.946
Mild DR	0.943
Moderate DR	0.948
Severe DR	0.939
Proliferative DR	0.940
Micro-average	**0.943**
Macro-average	**0.944**

The Severe DR class exhibits good class separation performance even with a rather smaller AUC. The micro-average and macro-average AUCs indicate that the classification ability of the model is balanced over all DR grades, so displaying high discriminative power free from bias toward any one grade.

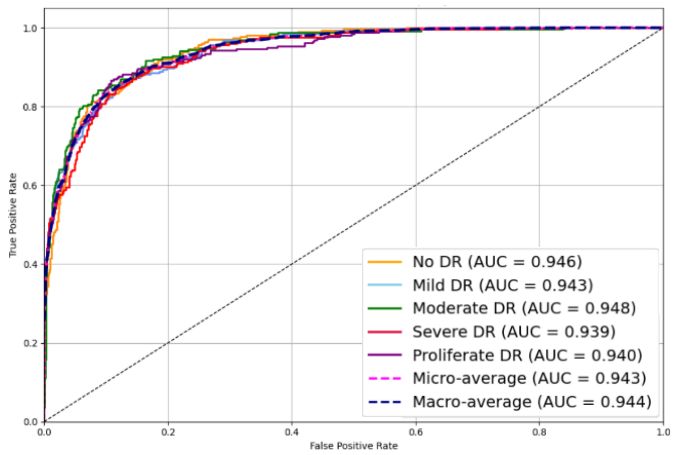

Fig. 7. Hybrid model – ROC-AUC Curve

The results show how effectively the model detects diabetic retinopathy and classifies among its several stages.

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F. Performance Comparison of the Proposed Model with Existing Models

The proposed solution outperformed all baseline models, with a maximum accuracy of 96.49% and superior diagnostic capabilities. Table VI compares the classification accuracy of different deep learning models used for DR detection.

TABLE VI. Performance Comparison of Proposed Model with Existing Models

Model	Accuracy %
VggNet19 [14]	82.17
CNN [15]	89
AlexNet [14]	89.75
GoogLeNet [16]	93.75
Inception [12]	94
Proposed Solution	**96.49**

In addition to higher accuracy, the proposed model exhibits greater consistency across all DR severity levels, making it more reliable for clinical use. Furthermore, its interpretable results and performance on augmented test data demonstrate its suitability for real-world use in a variety of healthcare settings.

V. CONCLUSION

The proposed hybrid model combines EfficientNet-B3, ResNet-50, DenseNet-121, and XGBoost for diabetic retinopathy classification. It delivers reliable and consistent performance across all severity levels. This provides doctors with a clear and dependable tool to make accurate and timely diagnoses. The model's performance may be limited by dataset availability and image quality differences across devices. Deployment in healthcare settings with limited resources may be restricted by the hybrid model's high computational requirements. Future research could focus on developing lightweight architectures that reduce computational demands, allowing for more widespread deployment in resource-constrained healthcare settings. Improvements like real-time detection with YOLO and lesion-level segmentation with U-Net or Mask R-CNN may increase the model's responsiveness and specificity, leading to improved retinal abnormality localization and a more efficient and accessible early diagnosis of diabetic retinopathy. Clinical integration with explainable AI to improve trust and usability should be prioritized.

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